

Measuring the Charge-diffusion Kernel of a Back-illuminated CCD in Orbit from Glancing Particle Tracks

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Abstract.

The lateral diffusion of photogenerated charge in the undepleted layer of a back-illuminated silicon sensor sets its point spread function and the statistics of its noise in the ultraviolet, but it is rarely measured on a flight sensor. We show that glancing particle tracks, which cross the full thickness of the sensor over a length of tens of pixels, sample the diffusion width at every depth in a single exposure, and we use them to measure the depth-dependent charge-diffusion kernel of the three CCDs of the Interface Region Imaging Spectrograph in orbit, from 1070 tracks found in level-1 images taken mostly inside the South Atlantic Anomaly. The width of the charge cloud at the back surface, the depth at which it vanishes, and the probability that two electrons deposited at the same depth are collected in the same pixel are measured directly and compared with a field-free diffusion model. On the Slit-Jaw Imager the same-pixel probability at the back surface is 0.34 ± 0.01 , against 0.34 from the model, and the tracks reveal a residual spread of about a micron inside the depletion region which the model neglects. The method needs no laboratory access and applies to any back-illuminated sensor in orbit.

Keywords: charge-coupled devices, charge diffusion, point spread function, ultraviolet detectors, particle tracks, IRIS.

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1 Introduction

A back-illuminated silicon charge-coupled device (CCD) absorbs ultraviolet (UV) light within a fraction of a micron of its back surface, in a layer where there is no electric field to sweep the photogenerated charge toward the gates. The charge diffuses until it reaches the edge of the depletion region, and the resulting charge cloud, which can be several microns wide, is shared among neighboring pixels. This sharing broadens the point spread function (PSF) of the sensor¹ and, because the electrons liberated by one photon are shared together, it changes the statistics of the image noise, entering the variance-to-mean ratio (VMR) of a photon-transfer curve through the probability that two electrons from the same photon land in the same pixel.²

The width of the charge cloud is set by the thickness of the undepleted layer, which depends on the resistivity of the wafer and the bias applied to the device, both of which vary between nominally identical sensors and are rarely known for a sensor in orbit. Laboratory measurements use light of varying penetration depth^{3,4} or the split events of soft X-rays,⁵ and neither is available once the instrument has launched.

Energetic particles offer a probe that arrives for free. A particle crossing the sensor at glancing incidence deposits charge along a track tens of pixels long, and because it enters at the back surface and exits at the gates, the fractional position along the track is the fractional depth of the charge. Cosmic-ray muons have been used in this way to measure the growth of charge diffusion with drift distance in thick, fully depleted CCDs,⁶ where the cloud widens with depth. In a thin sensor with an undepleted back surface the dependence is reversed, the cloud is widest at the back surface and collapses to a pixel-sharp line in the depletion region, and the same tracks measure the field-free kernel that governs UV imaging.

Here we apply this method to the three CCDs of the Interface Region Imaging Spectrograph (IRIS),⁷ whose far ultraviolet (FUV) and near-UV photon-transfer curves⁸ motivated the question. Section 2 describes the tracks, the model of their cross-section, and the fit; Section 3 presents the kernel of each sensor; and Section 4 compares it with the field-free model and discusses the systematics, including the charge density of the tracks.

2 Method

2.1 Tracks

We searched 3249 level-1 images from the IRIS spectrograph and the Slit-Jaw Imager (SJI), 680 of them taken inside the South Atlantic Anomaly (SAA), for straight streaks of charge against the dark background of off-limb or unexposed regions. The background of each frame was estimated as the trimmed mean of the frames in the same observing sequence, the read noise from the median absolute deviation of the residual, and pixels more than five times the read noise above the background were labeled. Connected groups of labeled pixels were fit with a straight line through the charge-weighted centroid of each row, and we kept the groups spanning at least twelve rows with a slope below 0.35 pixels per row, so that each track is nearly aligned with the pixel grid and its cross-section is sampled once per row. A seven-pixel-wide cutout centered on the fitted line was extracted for each track, which yielded 1070 tracks.

2.2 Model

We model the charge in every slice of a track as spread uniformly along the fitted centerline and diffused as a Gaussian with standard deviation

$$\sigma(t) = \sigma_{\max} \sqrt{1 - t/t_c}, \quad t < t_c, \quad (1)$$

and zero otherwise, where $t = z/D$ is the fractional depth of the slice below the back surface, $t_c = z_f/D$ is the fractional thickness of the field-free region, and $\sigma_{\max}$ is the width of the charge cloud at the back surface. A track that enters the back surface and exits the front surface spans the full thickness of the sensor, so the fractional depth of slice i of an N -slice track is $t = (i + 1/2)/N$, up to the orientation of the track, which we fit along with t_c and $\sigma_{\max}$ by exhaustive search on a grid of 21 values in each of $t_c \in [0, 1]$ and $\sigma_{\max} \in [0, 10] \mu\text{m}$. At each grid point the offset and tilt of the centerline are adjusted, by up to 0.6 pixels and 0.03 pixels per row, to minimize a robust misfit, $\sum \ln(1 + r^2/2)$, where r is the residual of the fraction of each slice's charge in each pixel in units of the read noise.

2.3 Selection

The useful tracks are those which cross the full thickness of the sensor at nearly constant energy loss. We kept only the tracks for which the fit constrains t_c to within 0.15 and improves on a model with no diffusion by at least ten units of misfit, and for which the median charge per slice in the last third of the track is within 50% of that in the first third, since a rise in the deposited charge along the track marks a particle that stopped inside the sensor. 417 tracks pass these cuts, 140 of them on the SJI CCD.

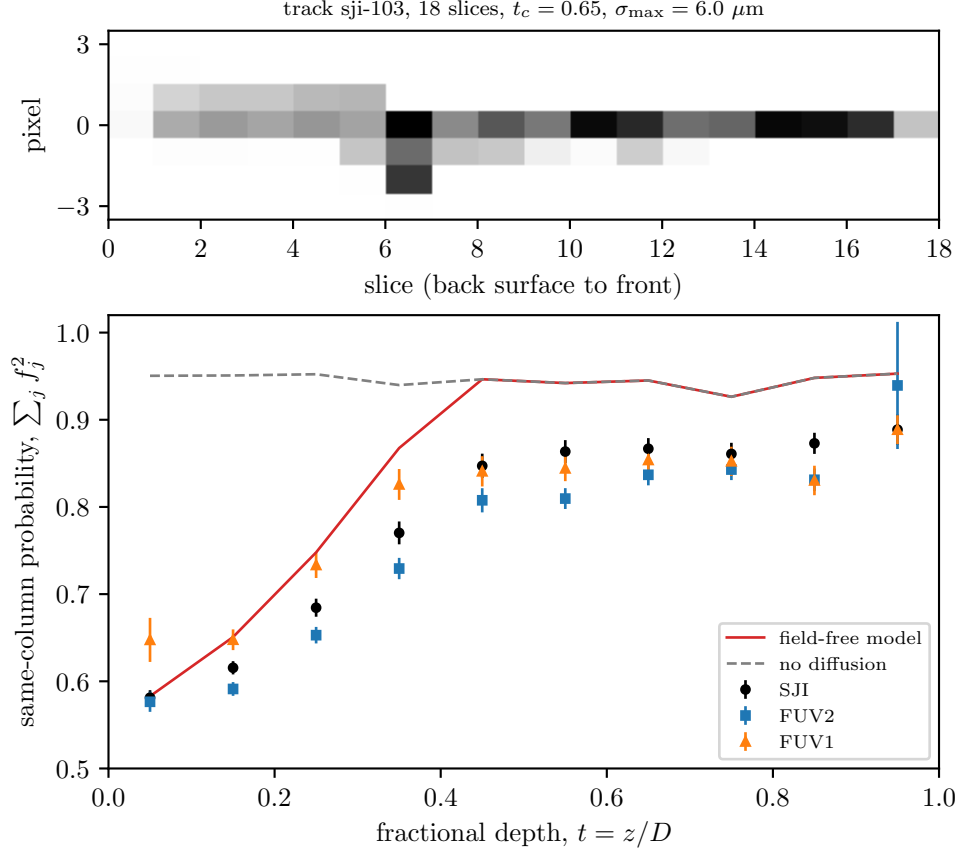


Fig 1 Top: a glancing particle track on the IRIS SJI CCD, in thousands of electrons per pixel, rotated so that it runs from the back surface (left) to the front surface (right). Bottom: the probability that two electrons deposited in the same slice are collected in the same column, averaged over the flat tracks on each CCD in bins of fractional depth, with the standard error of each mean. The solid line is the prediction of the field-free model, Equation 1 with $t_c = 0.38$ and $\sigma_{\max} = 5.34 \mu\text{m}$, evaluated at the fitted centerline of each SJI track and averaged in the same bins, and the dashed line is the same with no charge diffusion.

3 Results

Figure 1 shows one of the flat tracks along with the probability that two electrons deposited in the same slice are collected in the same column, $\sum_j f_j^2$ where f_j is the fraction of the slice's charge in column j , averaged over the flat tracks on each CCD in bins of fractional depth. This quantity requires no model of the shape of the kernel, and since the kernel is separable, the probability that two electrons are collected in the same pixel is its square. Table 1 lists the medians of the fitted parameters and the same-column and same-pixel probabilities for the slices within $D/10$ of the back surface, alongside the same-pixel probability predicted by the field-free model with $t_c = 0.38$ and $\sigma_{\max} = 5.34 \mu\text{m}$.

4 Discussion

On the SJI CCD the same-pixel probability at the back surface is 0.34 ± 0.01 , in agreement with the 0.34 predicted by the field-free model. Two features of the tracks deserve comment. First, the fitted t_c , with a median of 0.45 on the SJI CCD, is somewhat larger than the 0.38 of the model, and

Table 1 The charge diffusion measured from glancing particle tracks on the three IRIS CCDs. The number of frames is the number of level-1 images searched, with the number taken inside SAA in parentheses. The flat tracks are those which constrain t_c to within 0.15 and show no Bragg rise along their length. t_c and $\sigma_{\max}$ are the medians of the per-track fits, with the interquartile range in brackets. p is the probability that two electrons deposited within $D/10$ of the back surface are collected in the same column, $\mathcal{P} = p^2$ is the corresponding probability that they are collected in the same pixel, and the last column is $\mathcal{P}$ predicted by the field-free model for the same tracks.

CCD	frames	tracks	flat	t_c	$\sigma_{\max}$ (μm)	p	$\mathcal{P}$	$\mathcal{P}_{\text{model}}$
FUV1	2091 (211)	298	106	0.45 [0.35, 0.65]	4.5 [3.0, 5.5]	0.647 ± 0.025	0.419 ± 0.033	0.325
FUV2	2091 (211)	451	171	0.45 [0.40, 0.55]	5.5 [5.0, 6.5]	0.577 ± 0.012	0.332 ± 0.013	0.331
SJI	1158 (469)	321	140	0.45 [0.40, 0.50]	5.5 [5.0, 6.0]	0.581 ± 0.009	0.338 ± 0.010	0.340

beyond t_c the measured same-column probability settles below the value expected for a track with no diffusion at all. Both indicate that the charge undergoes some additional spreading, of order a micron, while drifting across the depletion region, which the model neglects. Second, the tracks are left by protons that deposit thousands of electrons per row, whereas a UV photon liberates one to three; whether the charge density of the track widens the cloud, as it does for slow protons in thick sensors,⁹ is a systematic that the tracks themselves can test by comparing the kernels of faint and bright tracks.

5 Conclusion

Glancing particle tracks measure the depth-dependent charge-diffusion kernel of a back-illuminated CCD in orbit, with no laboratory access, and on the IRIS CCDs they confirm the field-free model at the back surface while revealing a small residual spread inside the depletion region.

Disclosures

The authors declare that there are no financial interests, commercial affiliations, or other potential conflicts of interest that could have influenced the objectivity of this research or the writing of this paper. The analysis code and the text of this article were drafted with the assistance of Claude, a large language model, under the direction and review of the authors.

Code, Data, and Materials Availability

This article is executable:¹⁰ every figure, table, and number quoted in the text is computed from the tracks when the article is built. The code, the track cutouts, and the fits are available at <https://github.com/roytsmart/ccd-diffusion-paper>, and the level-1 IRIS images they were extracted from are available from the IRIS data archive.

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